

Haotian WEI

Ph.D. Candidate, 7+ yrs Theoretical Quantum Scientist

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EDUCATION

Ph. D. in Physics , Rice University, <i>Houston, TX</i>	Exp. Aug 2026
M. Sc. in Physics , Rice University, <i>Houston, TX</i>	Dec 2024
B.Sc. in Physics , Fudan University, <i>Shanghai, China</i>	Jun 2020
• Outstanding Graduate of Class 2020	
Visiting Student , University of California, <i>Berkeley, CA</i>	Aug - Dec 2017

PUBLICATIONS

Highlights: 8 papers | 320+ citations | 70+ media coverages | 3 Editors' Suggestions | Nat. Phys., PRL, PRXQ, Sci. Adv.

Google Scholar: scholar.google.com/citations?user=mu--7-UAAAAJ

- **Haotian Wei** and Kaden R. A. Hazzard. "A universal and efficient hybrid digital-analog fermionic quantum simulator" arXiv:2606.05517 (2026).
- Zheyuan Li, Rupsa De, Rishi Sivakumar, William Huie, **Haotian Wei**, Justin D. Piel, Chris H. Greene, Kaden R. A. Hazzard, Zoe Z. Yan, Jacob P. Covey. "Quantum science with arrays of metastable helium-3 atoms" arXiv:2601.06763 (2026). [*PRX Quantum Accepted*]
- Dasom Kim, Sohail Dasgupta, Xiaoxuan Ma, Joong-Mok Park, **Haotian Wei**, Liang Luo, Jacques Doumani, Xinwei Li, Wanting Yang, Di Cheng, Richard H. J. Kim, Henry O. Everitt, Shojiro Kimura, Hiroyuki Nojiri, Jigang Wang, Shixun Cao, Motoaki Bamba, Kaden R. A. Hazzard, and Junichiro Kono. "Observation of the magnonic Dicke superradiant phase transition" *Science Advances* **11**, eadt1691 (2025). [*Featured in news*]
- **Haotian Wei**, Eduardo Ibarra-García-Padilla, Michael L. Wall, and Kaden R. A. Hazzard. "Hubbard Parameters for Programmable Tweezer Arrays" *Physical Review A* **109**, 013318 (2024). [*Editors' Suggestion*]
- Zoe Z. Yan, Benjamin M. Spar, Max L. Prichard, Sungjae Chi, **Haotian Wei**, Eduardo Ibarra-García-Padilla, Kaden R. A. Hazzard, and Waseem S. Bakr. "Two-Dimensional Programmable Tweezer Arrays of Fermions" *Physical Review Letters* **129**.123201 (2022). [*Editors' Suggestion*] [*Featured in news*]
- Shintaro Taie, Eduardo Ibarra-García-Padilla, Naoki Nishizawa, Yosuke Takasu, Yoshihito Kuno, **Haotian Wei**, Richard T. Scalettar, Kaden R. A. Hazzard, and Yoshiro Takahashi. "Observation of Antiferromagnetic Correlations in an Ultracold SU(N) Hubbard Model" *Nature Physics* **18**.1356–61 (2022). [*Featured in news*]
- Eduardo Ibarra-García-Padilla, Sohail Dasgupta, **Haotian Wei**, Shintaro Taie, Yoshiro Takahashi, Richard T. Scalettar, and Kaden R. A. Hazzard. "Universal Thermodynamics of an SU(N) Fermi-Hubbard Model" *Physical Review A* **104**.043316 (2021). [*Editors' Suggestion*]
- Yuan D. Liao, Han Li, Zheng Yan, **Haotian Wei**, Wei Li, Yang Qi, and Zi Y. Meng. "Phase Diagram of the Quantum Ising Model on a Triangular Lattice under External Field" *Physical Review B* **103**.104416 (2021).

CONFERENCE PRESENTATIONS & TALKS

Invited talks:

- "A universal and efficient hybrid analog-digital fermionic quantum simulator", *IFF-CSIC 2026, Virginia Tech 2026, URochester 2026*.

Oral & poster presentations:

- "A universal and efficient fermionic variational quantum simulator", *APS Global Physics Summit 2026, ITAMP Winter School 2025, eQMA Spring School & Fall Workshop 2025, QuantIPS 2025, APS DAMOP Meeting 2025, Texas Quantum Summit 2025*.
- "Fermionic programmable quantum simulators running variational algorithms", *APS DAMOP Meeting 2024*.
- "Parameters and algorithms for programmable Fermi-Hubbard tweezer arrays for quantum simulations", *Rice Quantum Group Meeting 2023*.
- "Effective Hubbard parameters for programmable tweezer arrays", *APS March Meeting 2023*.
- "Programmable Hubbard model in tweezer arrays", *RCQM Workshop 2022, QuantIPS 2023*.
- "Hubbard parameters of optical tweezer arrays in arbitrary 1- and 2-D geometries", *IUPAP Conference on Computational Physics 2022*.
- "Stroboscopic fermion tweezer arrays: heating and Hubbard parameters", *APS DAMOP Meeting 2022*.

PROFESSIONAL SKILLS

Physics Techniques: Quantum Dynamics, Quantum Criticality, Fermi-Hubbard, RG, Variational Methods, Lie Algebra Representation Theory, Lieb-Robinson Bound, Mean Field Theory

Quantum Hardware: Neutral Atom, Ion Traps, Superconducting Qubits

Programming Languages: 7+ yr Python (NumPy/SciPy/PyTorch/Qiskit), 4+ yr C/C++, MATLAB, Mathematica & Julia

Algorithms: Variational Quantum Algorithms, Optimization, Tensor Networks, Matrix Diagonalization (full /Arnoldi/Lanczos), Monte Carlo, Numerical Analysis, Data Correlation Analysis, Regression, Deep Neural Networks

ML/DL: Coursera Deep Learning Specialization, Reinforcement Learning Course, neural-network design & training, LLM prompt design & engineering, regressions, SGD/Adam/BFGS optimizers, statistics (SSR/SST/Pearson R), SQL

Application Skills: High-performance computing (HPC), Git workflow, data visualization (Matplotlib/Inkscape), MySQL

Open-Source Contributions: GitHub profile github.com/htwei17

- *Pymanopt* (800+★) – extended core functionality of Riemannian manifold optimizer
- *HubbardTweezer* – individually developed gate-engineering toolkit of next-gen neutral-atom quantum hardware
- *Rice-thesis-latex* – restructured and modernized the most popular Rice University thesis LaTeX template

Languages: English (fluent), Chinese Mandarin (native)

Interests: Linguistics (phonetics/phonology), Saxophone

LEADERSHIP & OUTREACH

Physics & Astronomy Graduate Student Association (PAGSA), Rice University, Houston, TX Jun 2022 – Jun 2024

- Treasurer & Journal club coordinator

Academic Reviewer, Physical Review X, Physical Review Letters, Physical Review Research, Physical Review Applied, Physical Review A&B, European Physics Journal Plus Aug 2024 - Present

- Reviewed 20+ top research articles

APS March Meeting 2023, Las Vegas, NV Mar 2023

- Chaired a session in the world's largest physics conference with 14,000+ attendees